

SELF-STUDY GUIDE

TCS NQT Coding Practice

A structured revision manual for C++ problem solving, topic coverage, timed practice, and final-week exam readiness.

Built from 72 practice solutions in this workspace | 2026-05-21

How to use this guide

Treat every solved file as a memory-recall exercise. Read the problem, hide the solution, write the algorithm in five lines, code it under time, then compare with the reference solution only after testing your own output.

Current Material Snapshot

Difficulty	Available problems
Easy	47
Hard	7
Medium	18

Recommended Study Loop

1. Preview: read only the comment block and identify the core pattern.
2. Attempt: code from scratch with a timer and no reference.
3. Compare: inspect differences against the provided solution.
4. Stress-test: run minimum, maximum, duplicate, zero, and invalid-style inputs where relevant.
5. Redo: repeat the same problem after 48 hours without opening the old code.

30-Day Study Plan

Operating rule

Do not measure study by files opened. Measure it by problems solved without looking, mistakes logged, and repeated after two days.

Window	Focus	Daily output
Days 1-5	Arrays, loops, input/output discipline	2 easy + 1 medium daily; write dry-runs for edge cases.
Days 6-10	Strings, counting, hashing, validation	3 string problems daily; include one custom failing test.
Days 11-15	Math, number theory, series, patterns	Memorize formulas only after deriving them once.

Window	Focus	Daily output
Days 16-20	Sorting, two pointers, binary-search logic	Focus on invariants and boundary cases.
Days 21-25	Dynamic programming and graph traversal	Write state definition, transition, base case before code.
Days 26-30	Mixed timed mocks and revision	Solve 4 problems in 90 minutes; revise every logged mistake.

Pattern Playbook

Arrays and frequency

- Start with the simplest pass: sum, count, min/max, frequency map, prefix sum, or two pointers.
- When the array is sorted or paired, look for index parity, binary search, or duplicate grouping.
- For move-zeroes and Dutch National Flag, maintain named regions and update them in one pass.

Strings and validation

- Decide whether order matters. If order does not matter, use counts; if it does, scan with a window or stack.
- For validation problems, separate rule checking from transformation logic.
- Cast characters to unsigned char before ctype functions in C++ to avoid undefined behavior.

Math and series

- Write the first five terms manually, then identify whether the pattern is arithmetic, geometric, alternating, digit-based, or mixed.
- Prefer integer formulas over loops when the relationship is clear and safe.
- Check overflow whenever multiplication, factorial-style growth, or pair counts appear.

Dynamic programming

- State definition comes first: what does $dp[i]$ or $dp[s]$ mean after processing part of the input?
- For subset and coin problems, decide whether order matters; that choice controls loop direction.
- Base cases should be explicit before the transition is written.

Graph traversal

- Build adjacency carefully, mark visited when pushing into the queue, and sort neighbors only when output order is required.
- For cycle or reachability questions, track the exact condition that proves success or failure.

Problem-Solving Checklist

- [] Restate input and output in one sentence.
- [] Classify the problem: traversal, hashing, sorting, binary search, DP, graph, math, or string parsing.
- [] Write two sample tests and one edge test before coding.

[] Estimate time and space complexity before submitting.

[] After solving, write the mistake or key idea in one line.

Final-Week Revision Rules

- Redo all hard problems without looking at old code.
- Group mistakes by cause: wrong formula, off-by-one, input parsing, missed edge case, or inefficient approach.
- Run at least three mixed mocks. Stop after the timer and review honestly.
- Keep C++ templates minimal: fast I/O, vector, map/unordered_map, queue, and common helpers only.

Topic Coverage Map

Topic	Count	Problem numbers
Array	22	01, 02, 03, 06, 12, 19, 20, 22...
String	12	08, 09, 10, 29, 30, 31, 36, 40...
Math	26	11, 13, 15, 16, 17, 18, 21, 23...
Dynamic Programming	9	02, 05, 23, 28, 32, 35, 36, 37...
Graph	1	68
Bit Manipulation	4	06, 43, 61, 64
Binary Search	1	01
Sorting	1	20
Validation	1	40
BFS	1	68
Arrays	1	45
Calendar	1	63
Cipher	2	40, 46
Combinatorics	3	02, 04, 05
Greedy	1	58
Hash Map	3	50, 56, 59
Linked List	1	38
Maps	1	07
Mathematics	2	04, 05
Matrix	1	47
Number System	1	14
Pattern Printing	1	24
Physics	2	53, 57
Prefix Sum	2	41, 59
Prime	1	34
Series	3	18, 65, 70
Sieve	1	34
Simulation	5	07, 17, 51, 63, 66
Sliding Window	1	30
Two Pointers	3	03, 27, 71

Full Practice Index

No.	Problem	Level	Topics
01	Odd Occurring Element in $O(\log N)$	Hard	Array, Binary Search
02	Count Subsets with Given Sum	Hard	Array, Dynamic Programming, Combinatorics
03	Chocolate Packets — Move Empty Packets to End	Easy	Array, Two Pointers
04	COVID Handshake Problem	Easy	Mathematics, Combinatorics
05	Book Exchange — Derangement Count	Hard	Mathematics, Dynamic Programming, Combinatorics
06	XOR Coin Denomination	Easy	Bit Manipulation, Array
07	Income, Expenditure and Savings Tracker	Medium	Simulation, Maps
08	Reverse Words in a Sentence	Easy	String
09	Count Vowels, Consonants, Spaces and Special Characters	Easy	String
10	Check Palindrome (Case-Insensitive)	Easy	String
11	Armstrong Number Check	Easy	Math
12	Second Largest Number in Array	Easy	Array
13	Leap Year Check	Easy	Math
14	Binary Number Conversion	Easy	Number System
15	GCD and LCM	Easy	Math
16	Digit Sum Until Single Digit	Easy	Math
17	Average Sales Calculation	Easy	Math, Simulation
18	Number Pattern Series (Interleaved A and B)	Easy	Math, Series
19	Smallest Number in Array	Easy	Array
20	Sorting Algorithm (Bubble Sort)	Easy	Sorting, Array
21	Prime Number Check	Easy	Math
22	Remove Duplicates from Array	Easy	Array
23	Fibonacci Series — Nth Term	Easy	Dynamic Programming, Math
24	Pattern: Right-Angle Triangle of Numbers	Easy	Pattern Printing
25	Rotate Array Left by K Positions	Medium	Array
26	Find Missing Number in 1 to $N+1$	Easy	Array, Math
27	Sort Array of 0s, 1s and 2s (Dutch National Flag)	Medium	Array, Two Pointers
28	Maximum Subarray Sum (Kadane's Algorithm)	Medium	Array, Dynamic Programming
29	Check if Two Strings are Anagrams	Easy	String

No.	Problem	Level	Topics
30	Longest Substring Without Repeating Characters	Medium	String, Sliding Window
31	Find Maximum Occurring Character	Easy	String
32	Pascal's Triangle	Medium	Math, Dynamic Programming
33	Sum of Digits of a Number	Easy	Math
34	Sieve of Eratosthenes — All Primes up to N	Medium	Math, Prime, Sieve
35	Subset Sum Problem (DP)	Hard	Array, Dynamic Programming
36	Longest Common Subsequence (LCS)	Hard	Dynamic Programming, String
37	Minimum Coin Change	Hard	Dynamic Programming
38	Cycle Detection in Linked List (Floyd's Algorithm)	Medium	Linked List
39	Closest Multiple of Y to X	Easy	Math
40	Password Validator + Caesar Cipher	Medium	String, Validation, Cipher
41	All Subarrays with Sum Equal to K	Medium	Array, Prefix Sum
42	Fibonacci Sum up to N Terms	Easy	Math, Dynamic Programming
43	Distinct Bitwise OR of Subarrays	Hard	Bit Manipulation, Array
44	Two-Wheeler and Four-Wheeler Production	Easy	Math
45	MPCS Fitness Test — Average Oxygen Level of Trainees	Medium	Arrays
46	Caesar Cipher (With Digits)	Medium	String, Cipher
47	Matrix Addition	Easy	Matrix
48	Reverse a String Without Built-In Functions	Easy	String
49	Check if Number is Perfect	Easy	Math
50	Count Frequency of Each Element in Array	Easy	Array, Hash Map
51	Weekly Exercise Summary	Easy	Simulation
52	Count Palindrome Numbers in Range M to N	Easy	Math
53	Train Travel Time on Bridge	Easy	Math, Physics
54	Split Array with Equal Averages	Medium	Array, Math
55	Perfect Donation Amount	Easy	Math
56	Inventory Frequency Counter	Easy	String, Hash Map
57	Calculate Speed from Distance and Time	Easy	Math, Physics
58	Jump Game — Can You Reach the End?	Medium	Array, Greedy
59	Count Subarrays with Sum = K	Medium	Array, Hash Map, Prefix Sum
60	First Non-Repeating Character	Easy	String

No.	Problem	Level	Topics
61	Toggle Bits — Number of Bits to Flip	Easy	Bit Manipulation
62	AP/GP Series: Find Nth Term	Easy	Math
63	Count Sundays in a Month	Easy	Simulation, Calendar
64	Toggle All Bits of a Number	Easy	Bit Manipulation
65	Number Series: $n^2 + n$ Pattern — Find Next Term	Easy	Math, Series
66	Alice's Verbal Kho-Kho — Digit Passing Game	Medium	Math, Simulation
67	Distinct Elements in Array	Easy	Array
68	BFS (Breadth First Search) on a Graph	Medium	Graph, BFS
69	Factory Production (Rate Calculation)	Easy	Math
70	Nth Term of Mixed Series $A(n)=n$ and $B(n)=n^2$	Easy	Math, Series
71	Sort 0s, 1s and 2s — Dutch National Flag (2026 Pattern)	Medium	Array, Two Pointers
72	Move Zeroes to End — Chocolate Factory (2026)	Easy	Array

Daily Practice Tracker

Day	Problems solved	Main mistake	Redo date	Status
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
Quality bar A problem is complete only when you can explain the approach, code it again from memory, and name the edge case that would break a careless solution.				